

The IS-LM-PC Model

Intermediate Macroeconomics

Fall 2025, Sciences Po Campus du Havre

Lecture 10

Announcements

- Book Chapter 9
- Final Exam: projected date **December 3, during your normal class time**
- Quiz 4 posted today, due on Wed 11/24 at 11:59PM

Building blocks

- **IS-LM model** (short-run)

The short-run output is determined by demand (the Goods market)

$$IS : Y = C(Y - T) + I(Y, r + x) + G$$

$$LM : r = \bar{r}$$

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Relationship between unemployment and inflation

$$\pi - \pi^e = -\alpha(u - u_n)$$

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- Step 1: We need to formally establish a link between output and unemployment

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- Unemployment rate

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- As a result:

$$Y = L(1 - u)$$

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- It is the unemployment rate at which price and wage setting decision are consistent (recall the Labor market)
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- There is some natural level (associated with frictional and structural unemployment)

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Potential output (Y_n):

- Output level that occurs at the natural rate of unemployment

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Difference between actual and potential output:

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We can see that:

- $u > u_n \implies$ Output gap is negative
- $u < u_n \implies$ Output gap is positive
- $u = u_n \implies$ Output gap is zero, actual output = potential

Output Gap and the Phillips curve

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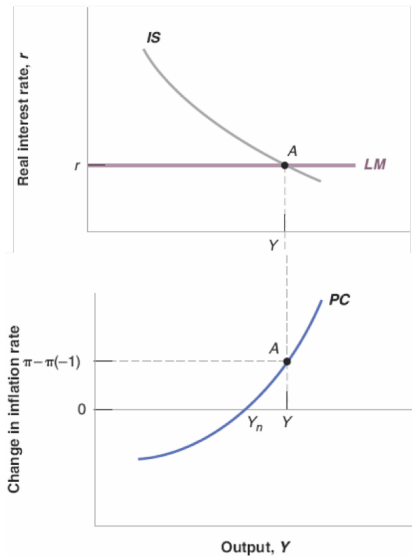
Final simplification: $\pi^e = \pi_{t-1}$

$$\begin{aligned}\text{Phillips Curve: } \pi - \pi_1 &= \frac{\alpha}{L}(Y - Y_n) \\ \Delta\pi &= \frac{\alpha}{L}(Y - Y_n)\end{aligned}$$

$$\Delta\pi = \frac{\alpha}{L}(Y - Y_n)$$

- $Y > Y_n \implies \Delta\pi > 0$: when output is above its potential, the inflation increases
The economy is overheating
- $Y < Y_n \implies \Delta\pi < 0$: when output is below its potential, the inflation falls
The economy is recessionary

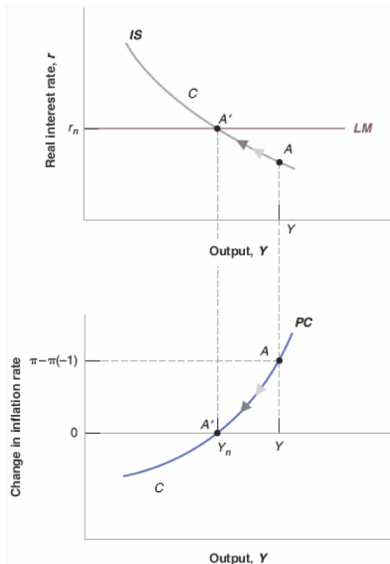
The IS-LM-PC model



IS-LM (Top graph): A low policy rate leads to higher output.

PC (Bottom graph): A higher output leads to a larger change in inflation.

The IS-LM-PC model and the Medium Run Equilibrium



- Over the medium run, the economy converges to the natural level of output and stable inflation.
- At the medium-run equilibrium (point A'), r_n is called the **natural, neutral, or Wicksellian rate of interest**.
- If the central bank wants to achieve a constant level of inflation, then the initial boom must be followed by a recession.

Dynamics and the Medium Run Equilibrium

- Assume instead that the expected inflation rate is a constant, then PC equation becomes:

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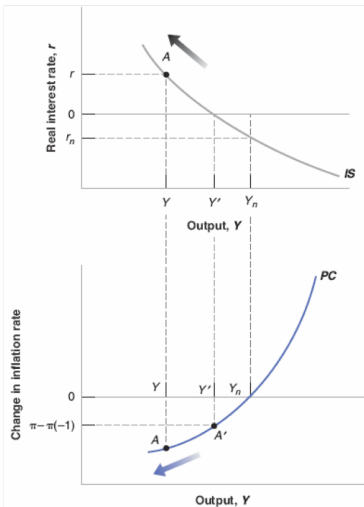
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- A positive output gap generates a higher level of inflation, rather than an increase in inflation.
- So long as inflation expectations remain **anchored**, the central bank does not need to compensate for the initial boom by a recession later.

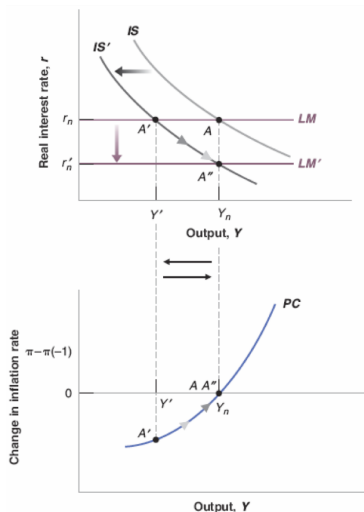
Extensions

1. ZLB and the Deflation Spiral



- The zero lower bound constraint may make it impossible to achieve a negative real policy rate.
- **Deflation spiral** or **deflation trap** occurs at Y' when output is still below potential, and thus inflation is still decreasing.
- Lower output leads to more deflation, and more deflation leads to a higher real interest rate and lower output.

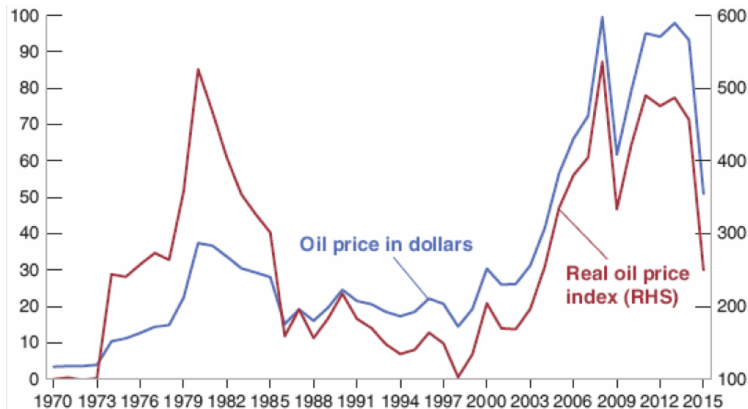
2. Fiscal Consolidation Revisited



- Fiscal consolidation leads to a decrease in output in the short run.
- In the medium run, output returns to potential, and the interest rate is lower.

3. The Effects of an Increase in the Price of Oil

Over the last 40 years, there have been two sharp increases in the real price of oil, the first in the 1970s and the second in the 2000s.



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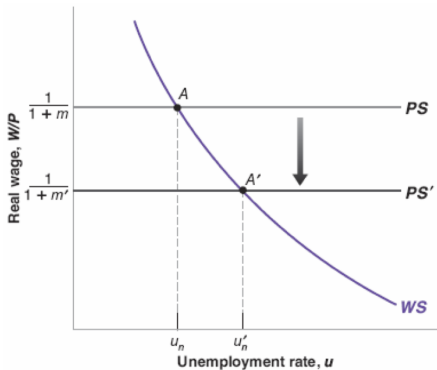
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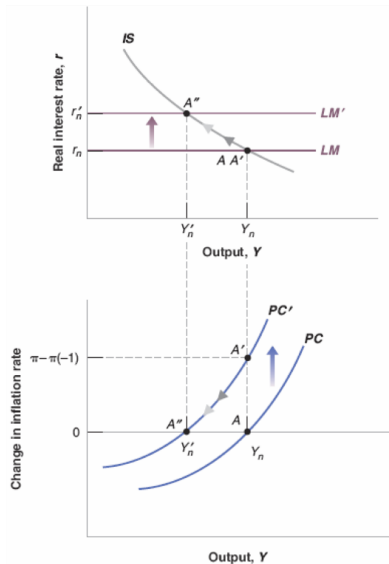
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- 2008: A large recession led to a sudden decrease in the demand for oil, and thus falling oil prices.
- 2014 and after: A combination of increased supply due to the increase in U.S. shale oil production and the partial breakdown of OPEC led to sudden drop in oil prices.

3. Oil Prices and Unemployment



- An increase in the price of oil is equivalent to an increase in the markup.
- It leads to lower real wages and a higher natural rate of unemployment.

3. Oil Prices and IS-LM-PC



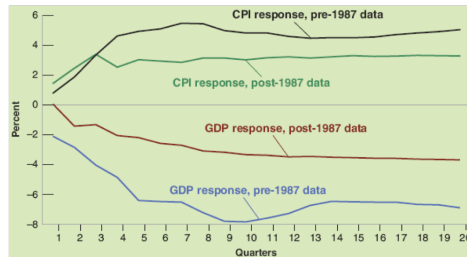
- If A' is the short-run equilibrium, then if the central bank increases the policy rate to stabilize inflation, then the economy moves to its medium-run equilibrium at point A'' .
- **Stagflation** (stagnation and inflation) occurs along the way.

3. Oil Price Increases: Why Were the 2000s So Different from the 1970s?

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The Effects of a 100% Permanent Increase in the Price of Oil on the CPI and on GDP

- The effects of an increase in the price of oil on output and the price level are smaller than they used to be.
- Plausible explanations include U.S. workers' lower bargaining power, and inflation that was more anchored by monetary policy.



Conclusions

- Shocks or changes in policy typically have different effects in the short run and in the medium run.
- Disagreements about the effects of various policies depend on how fast you think the economy adjusts to shocks.
- Movements in output around its trend are called output fluctuations (business cycles).
- Economic fluctuations are the results of shocks and their dynamic effects, called the propagation mechanism.